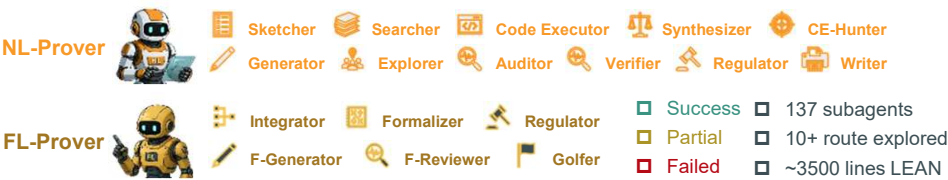
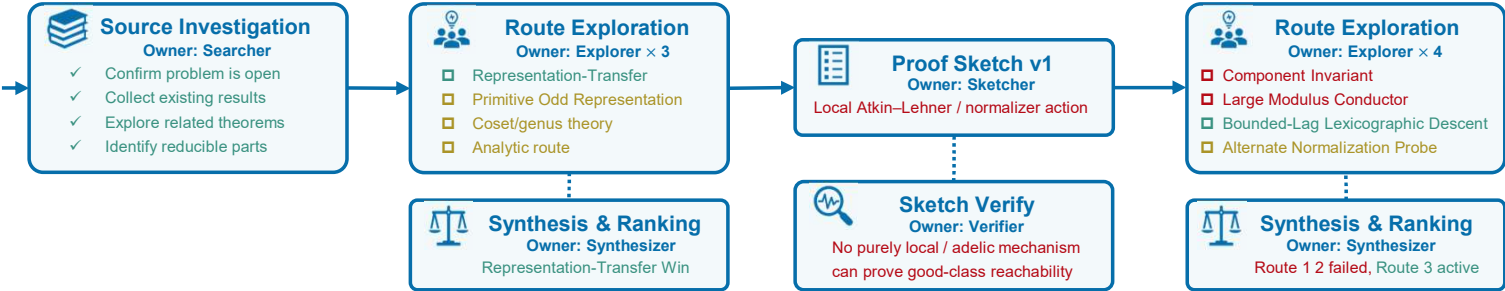


Prove that any nonnegative integer can be written in the form:

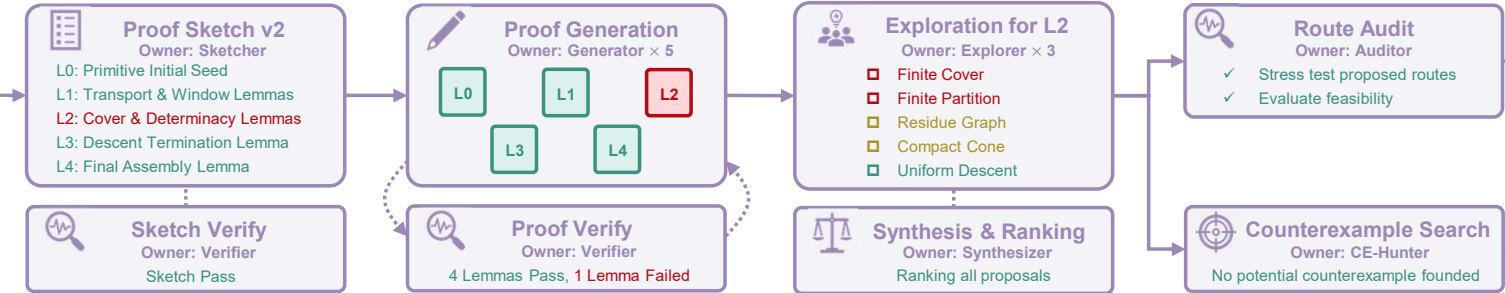
$$\frac{x(x+1)}{2} + \frac{y(3y+1)}{2} + \frac{z(5z+1)}{2}, \quad x, y, z \in \mathbb{N}$$



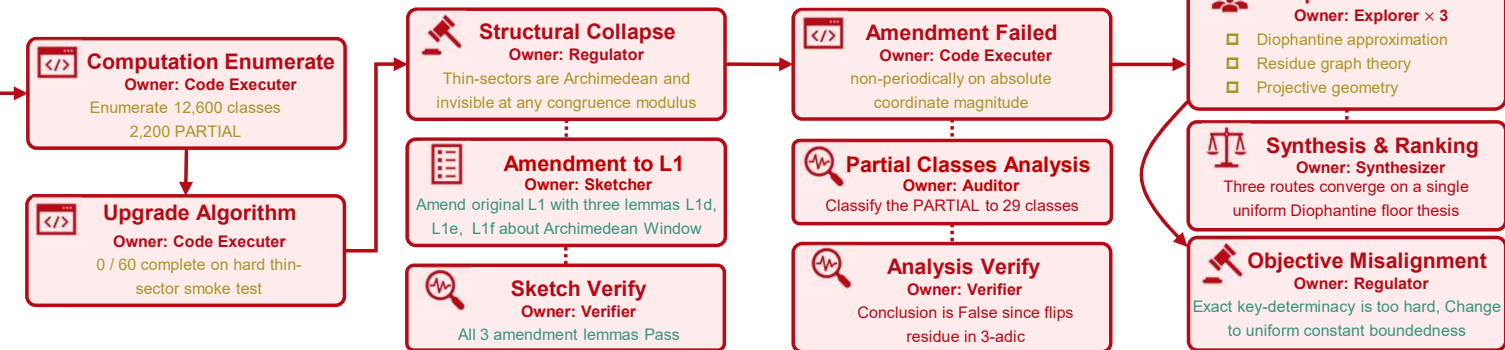
Problem Understanding and Route Exploration



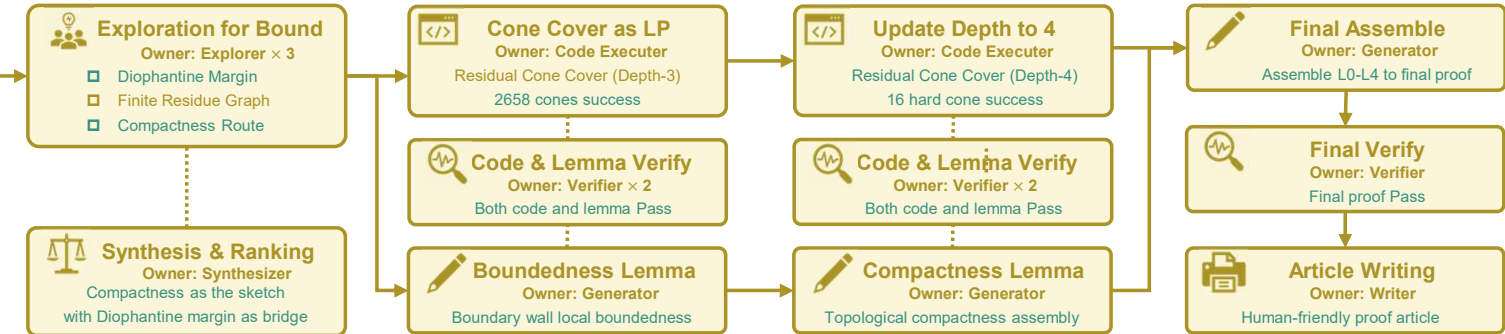
Proof Development and Bottleneck Identification



Mechanism Blockage and Strategic Recovery



Computation Certification and Proof Assemble



LEAN 4 Formalization and Verification

